Nicholas P. Dana, PhD

Entrepreneur, Collaborator, AI & Cross-Industry Innovator

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SUMMARY (Updated Jan. 2026)

Innovative data scientist and AI engineer integrating advanced analytics and AI to accelerate scientific discovery and product innovation. Proven leader in building scalable systems, automating complex pipelines, and translating R&D breakthroughs into operational tools that drive value across business applications and technology sectors.

EXPERIENCE

Elephas Biosciences — *Principle AI Scientist & Developer*

June 2021 – Present | Remote / Madison, WI

- Architected and deployed **multi-modal analytics pipelines** in cloud-integrated ML workflows leveraging image, cytokine, and assay data to predict drug response in *ex vivo* tumor models.
- Co-authored multiple conference & BioRxiv publications advancing translational oncology (**see below**).
- **Leadership Impact**: Technical lead for SME group (8) developing custom analytics achieving **>96% training data accuracy** and **100% concordance** with validation samples.
- **Strategic Initiative**: Prototyped analytic methods leading to company's first lab derived test offering.
- **Strategic Initiative**: Led SME group (6) identifying key QA/QC metrics, significantly improving QNS assessment and overall platform accuracy

GeoVisual Analytics — *Computer Vision & Machine Learning Scientist*

2018 – 2021 | Boulder, CO

- Built and trained **deep-learning CV models** for agricultural analytics using UAV imagery.
- Developed crop maturation and yield forecast models integrating image, weather, and GIS data.
- **Quantifiable Result**: Achieved **0-day median error** in harvest date prediction and **<5% error** in crop yield estimation across production sites.

Essen BioScience (Sartorius) — *Algorithm Engineer*

2016 – 2018 | Ann Arbor, MI

- Designed and implemented segmentation and denoising algorithms enabling automated live-cell image analysis with minimal user input.
- **Quantifiable Result**: Developed a computer vision library for time-series imaging, achieving **99.5% data fidelity** at **14× probabalistic compression efficiency**.

CERTIFICATIONS & CREDENTIALS

Post Graduate Program in AI Agents for Business Applications

University of Texas at Austin, McCombs School of Business

In Progress - Completion Q2 2026

Accelerated course giving students the technical and conceptual depth to design, deploy and govern Agentic AI solutions for a variety of business applications

EDUCATION

Doctor of Philosophy, Biomedical Engineering

The University of Texas at Austin, 2016

Dissertation: *Photoacoustic Imaging for Tissue Characterization and Image Guidance in Cardiovascular Applications*

Bachelor of Science, Physics

University of Utah, 2010

Focus in Medical Physics, Mathematics and Chemistry

SELECTED PUBLICATIONS

- Dana N. *et al.*, **"A cytokine-based classifier to predict response to immunotherapy using an *ex vivo* live tumor fragment platform"** [Working title, Manuscript in preparation].
- Dana N. *et al.*, **"Development and characterization of cytokine-based classifiers to predict response to immunotherapy using an *ex vivo* live tumor fragment platform"**, *SITC 2025 Poster*.
- Ramasubramanian T.S. *et al.*, **"A live tumor fragment platform to assess immunotherapy response in core needle biopsies,"** *bioRxiv*, July 2025.
- Dana N. *et al.*, **"Optimization of dual-wavelength intravascular photoacoustic imaging of atherosclerotic plaques using Monte Carlo optical modeling"** *JBO* 2017.
- Dana N. *et al.*, **"Overview of Photoacoustic Imaging"**, *Advances in Medical Physics*, 2016 edition.

TECHNICAL SKILLS

Languages: Python, SQL, C++, C#, R, MATLAB, Docker

AI / Data Science: GenAI, AgenticAI, LangChain / LangGraph, PyTorch / Tensorflow, Scikit, Pandas, Bayesian Methods, Dimensionality Reduction, Clustering, Computer Vision

Systems & Tools: Linux, Git, CI/CD, Cloud-based container orchestration

Domain Expertise: Optical imaging, assay analytics, high-dimensional analysis, bioinformatics, agricultural AI, and computational modeling